

Evolving Battery Management Systems for EV

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Abstract— In the new age of technology, we face different problems such as an increase in global warming and pollution, depletion of conventional energy resources, and carbon footprint. This leads to the use of electric vehicles. Choosing an energy storage system for these new technologies is an important task. So battery technology can be a good solution to the question. Hence, the battery technology plays an important role in energy storage in these areas. It satisfies the purpose of being portable with a small size and can be placed anywhere. The performance of the battery depends on various factors like chemistry, temperature variation tolerance, load profile, and charging-discharging algorithm. For monitoring and controlling the battery performance, a battery management system is established. It requires estimation of different parameters for the choice, monitoring, and controlling of the battery for better performance and efficiency when used for an electric vehicle. The paper covers topics related to battery parameter importance, estimations, and battery management systems for efficient operation. Common observations are like battery SOC estimation, SOH estimation, and ST estimation for an efficient EMS.

Keywords—Battery Management System, EV Battery System, Li-ion battery for EV

I. INTRODUCTION

In the view of the advancement of all areas to achieve the global market, transportation is the key to spreading worldwide. Road transport is a basic, easily available, and economical mode of transportation. However, the rapid expansion of transportation systems has led to environmental issues such as noise pollution, air pollution, and global warming. These problems contribute to carbon footprints, ozone layer depletion, and the depletion of conventional energy sources. It leads to the degradation of a healthy life. The solution to these problems leads towards the new technology, which introduces the use of non-conventional sources, an environmentally friendly transportation system. An electric vehicle could offer a solution to the issues mentioned earlier. Along with the transportation, some new technologies are also emerging in small portable devices used for many applications. The main system for all these devices is to supply electrical energy[1].

A. Importance of energy storage devices for EV

The conventional transportation devices use an IC engine for propulsion, which uses fuels such as diesel, petrol. The infrastructure for refueling these vehicles is available, which increases convenience to use IC engine vehicles. The electric vehicle uses electrical energy for propulsion. The electrical energy storage device offers a solution that ensures convenience and flexibility, improves grid reliability and power quality, and supports the expansion of renewable energy integration. There are different energy storing devices available, such as battery, fuel cell, super capacitor,

flywheels, etc. But the battery plays an important role in energy storage in these areas as it satisfies the purpose as a portable device with a small size and can be placed anywhere.

Many types of rechargeable batteries are available in the market but Li-ion batteries are highly preferred in the electric vehicle technology. These batteries possess characteristics better than other types of batteries.

As per the recent studies it is found that, though battery technology has good performance, it has some disadvantages like short life time, fast degradation rate, risk of getting flames. The working of the battery depends on chemical process within it while charging and discharging. Its operation is non-linear as it is influenced by multiple factors, including temperature variation, battery chemistry, connected load profile, charging-discharging algorithm. Hence a comprehensive research is focused on the management of charging & discharging of batteries, charge equalization, temperature management, and estimation of various parameters to control performance degradation of the battery.

B. Importance of BMS for EV

Battery is a DC device which transforms chemical energy into electrical energy. The charging-discharging of the battery within the limits of parameter increases life of battery. So, to have efficient use, battery needs continuous monitoring of the parameters such as battery voltage, current and temperature[2]. The system is called Battery Management System (BMS) which does all task to improve performance of battery. So there are several tasks done by BMS. The battery management system is also doing functions like estimation and monitoring of voltage, current, temperature, power management, cell monitoring, temperature control, heat management, fault analysis, battery protection, communication and networking[3]. The BMS plays a crucial role in monitoring the battery's real-time status and accurately estimating the remaining energy available for delivery to the load. Additionally, it extends battery life by addressing charge imbalance issues in series-connected cells[4-5]. So, BMS analyses battery parameters to enhance battery health management and minimize the need for frequent battery replacements.

To ensure safety, reliability and performance BMS standards are introduced. Standards such as ISO 26262 (functional safety), UNECE R100 (electric vehicle safety), and IEC 62660, 61508, 62619 (battery safety and performance) are commonly followed in the automotive industry. By defining requirements for thermal management, overvoltage protection and fault diagnosis, these standards ensure safe operation of EV. Compliance with these standards is essential for certification and market approval,

promoting consumer trust and supporting the widespread adoption of electric vehicles.

II. BATTERY TYPES

A variety of batteries are available on the market, including lead-acid batteries, nickel-based batteries, nickel-metal hydride batteries, lithium-ion batteries, solid-state batteries, high-temperature batteries, and flow batteries. Among all these options, the Li-ion battery is the most suitable solution for the application as an Electric Vehicle. Lithium-ion batteries offer several advantages, including temperature sensitivity, compact volume, low maintenance requirements, recyclability, lightweight design, and long lifecycle. By considering electrical aspects, Li-ion batteries have good energy storing capacity, high specific power and energy, low loading effect & high efficiency[6]. They also exhibit excellent charge-discharge efficiency in converter-based energy-storing systems, along with high energy density and power rating. However, they are sensitive to charging and discharging conditions, making them prone to issues such as deep discharge or overcharge, which can negatively impact performance and lifespan can also lead to hazardous conditions like fire[1].

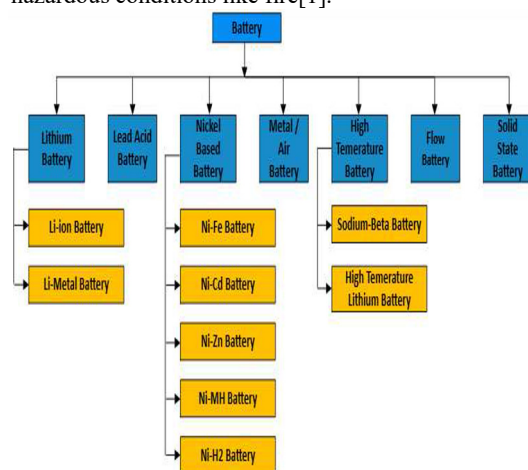


Fig. 1. Classification of Battery

III. BATTERY MANAGEMENT SYSTEM (BMS)

The battery-operated electric vehicles or plug-in hybrid electric vehicles need high current and voltage. To deliver high current and high voltage, a battery pack is composed of multiple cells connected in both series and parallel configurations[7]. The combination can be 1S2P, 1S3P, and 2S2P[1]. The number of cells makes one module, and a combination of several modules makes a battery pack. So there is a requirement to monitor pack parameters, module parameters & cell parameters separately.

The basic structure of the battery management system is divided into 3 parts as CMU (cell monitoring unit), MMU (module monitoring unit), and PMU (pack monitoring unit) [3]. BMS gathers all data from the sensors to calculate parameters such as state of charge (SOC), state of health (SOH), estimation of remaining useful life (RUL), State of Safety (SOS), State of Temperature (SOT), State of Power (SOP) and State of Energy (SOE) of the battery[1],[8]. The main work of BMS is to execute all the algorithms with

measured and calculated data as input information and send a command to control the battery function [2],[9].

Battery Management Systems (BMS) can be developed as integrated ICs or customized based on specific requirements. They are classified into three types based on topology: Modular BMS, Centralized BMS, and Distributed BMS. Despite their different topologies, all three function similarly in terms of monitoring, control, and protection[6].

BMS components are mainly divided into two main categories: Hardware and software. The hardware part comprises Sensors and a Monitoring circuit, a charge-discharge circuit, a safety circuit, and data acquisition.

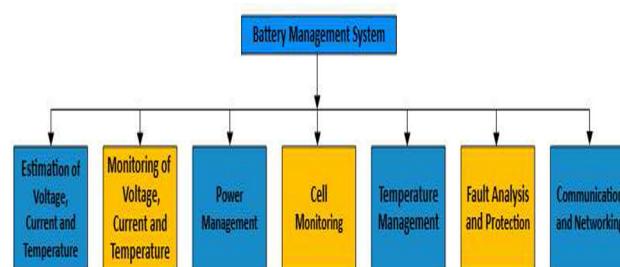


Fig. 2. BMS Function Block Diagram

The software part comprises of battery modeling state estimation, fault detection, and data storage. The components required to do the operation are:

- Sensors: To collect data of each cell to monitor voltage, current, and temperature
- Control unit: To execute the algorithms to estimate different parameters, make decisions, and send commands to other devices
- Communication devices: To send the control signals to peripheral devices to operate the charging and discharging circuit, relays, and contactors[10].
- Display: To display estimated parameters like SOC, SOH, and SOT, Charging & discharging status, and alarms for protection.

IV. ESTIMATION OF PARAMETERS

Recent studies show that lithium-ion batteries are the most effective option for ensuring the efficient performance of electric vehicle batteries. To overcome obstacles in the smooth operation of the battery, a battery management system (BMS) is used. A battery has several general specifications as nominal voltage, cut-off voltage, ampere-hour rating, C-rate, operating temperature range & and charging-discharging rate and charging-discharging cycles. The Battery Management System (BMS) uses various battery parameters for estimating some critical parameters to monitor, control, and protect the battery pack[10]. The critical parameter estimations are listed below:

A. State of Charge Estimation (SOC):

The state of charge of the battery represents the present capacity of the battery. SOC is related to lithium-ion transport between electrodes. A high SOC means a fully lithiated anode and a de-lithiated cathode.

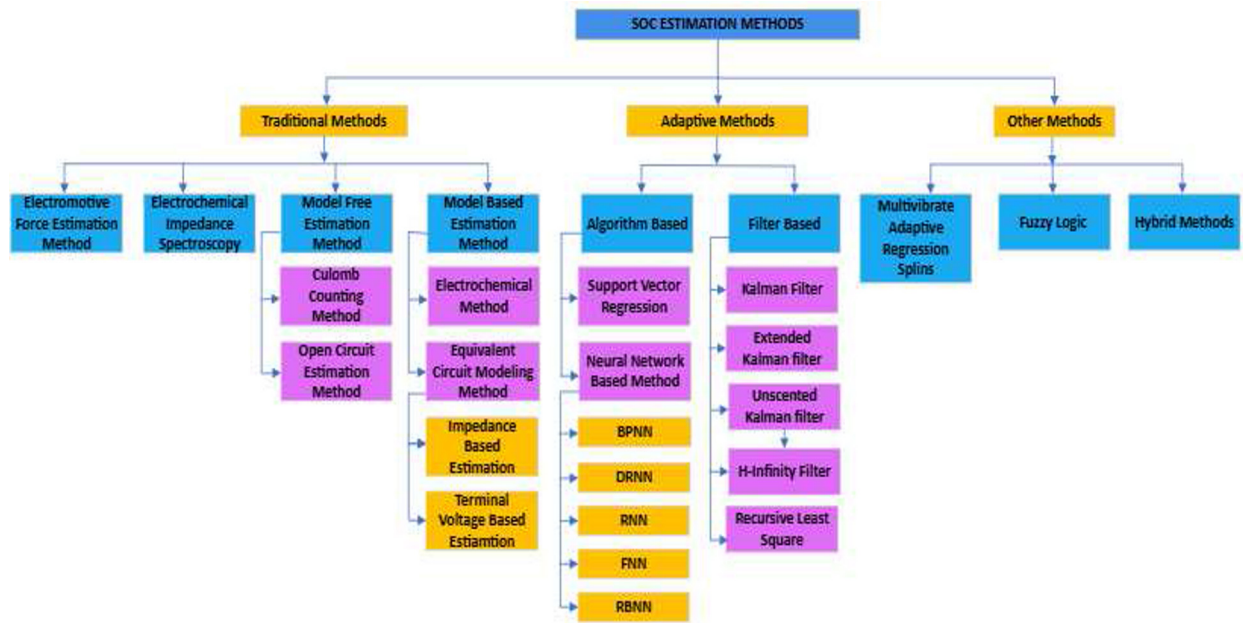


Fig. 3. Classification of Battery SOC Estimation Methods

It is the amount of capacity of the battery that remains after discharge from the highest charged condition. It is the instantaneous percentage of remaining charge after utilization. So is the most important indicator to make decisions about charging and discharging of a battery pack. The nominal capacity of a battery is the charge extracted between cut-off voltage limits. These voltage limits are typically 4.2V (upper) and 2.5V (lower) throughout battery life [11-14],[5].

The State of Charge (SOC) Estimation Methods for batteries are classified into three main groups. This classification highlights different strategies for estimating SOC in battery management systems, ranging from traditional, physics-based approaches to modern, data-driven, and hybrid methods.

- **Traditional method:** These methods use model-based and model-free methods. The main commonly used methods are the Coulomb counting method, open circuit estimation method, electrochemical model-based method, and equivalent circuit modeling method [11].
- **Adaptive Methods:** These methods use algorithm-based and filter-based approaches. It basically depends on data collection and model training mechanisms like neural networks, support vector regression. Also designing filters to refine estimates over time and handle noisy real-world data. These methods are useful for handling measurement noise, nonlinearity, uncertainty, and are adaptive to dynamic battery conditions [15-17].
- **Other Methods:** Multivariate Adaptive Regression Splines (MARS) use data-driven regression models to capture nonlinear SOC relationships, while Fuzzy Logic applies rule-based reasoning to handle uncertainties in battery behavior. Hybrid Methods enhance SOC estimation by combining techniques

like Kalman Filters, Neural Networks, and Model-Based approaches for improved accuracy and adaptability[11].

B. State of Health Estimation (SOH):

State of Health (SOH) calculation is a vital parameter in the implementation of a Battery Management System (BMS). It reflects the overall condition of the battery and its capability to store and deliver energy compared to a newly manufactured one. It is typically expressed as a percentage to its original capacity to deliver energy. So 100% SOH represents new battery and 0% SOH indicates complete degradation of the battery. The basic indicator in SOH estimation is the capacity of the battery. SOH, on the other hand, is defined as the ratio of maximum available capacity to the nominal capacity. The health of the battery is mainly influenced by capacity degradation, increase in internal resistance, cycle count, temperature effect, and aging. SOH estimation is used for assessing battery lifespan, efficiency, and performance, which leads to the changing or replacing battery packs in EVs for smooth operation[18].

The State of Health (SOH) estimation methods include electrochemical models, Equivalent circuit models (ECM), and data-driven (DD) algorithms.

- **Electrochemical model-based method:** It provides physical insights about the battery, such as ion concentrations, over potentials beyond the terminal voltage. These models help to predict degradation using estimation of variables directly linked to aging mechanisms[12].
- **Equivalent Circuit model-based method:** It uses circuit elements like resistors and capacitors to approximate battery behavior. It has a simple structure. Their parameter depends on SOC and current, but has limited physical meaning.

- Data-driven algorithm-based method: DD algorithms extract relationships between inputs and outputs without considering battery chemistry. They are commonly utilized for simultaneous SOC and SOH estimation, taking into account factors such as ion loss, degradation of active materials, and increased resistance.

Electrochemical models provide the most detailed insights, ECMs offer simplicity, and data-driven models leverage large datasets. A hybrid approach combining these methods could improve battery state estimation accuracy [15],[17],[19-21].

C. State of Temperature Estimation (SOT):

State of Temperature (SOT) is the temperature analysis of a battery that includes the surface and internal temperature of the battery. Surface temperature is the temperature outside the battery, and internal temperature is higher than which true heat of the battery. The variation in the temperature affects the performance and safety of the battery. The figure below shows the different effects of temperature on the battery.

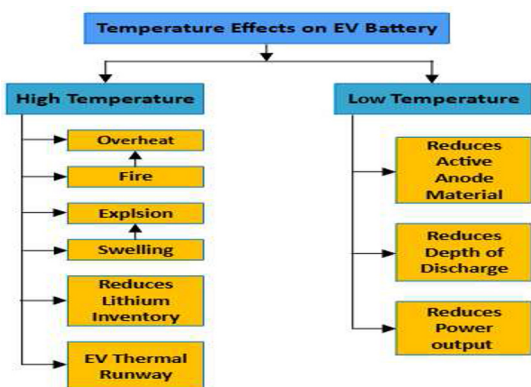


Fig. 4. Temperature effects on the battery

The maintenance of uniform battery temperature is difficult because it depends on a thermal management strategy. There is uneven cooling among cells, heat is generated mainly in the core of the battery pack. So surface cooling increases the temperature difference between the outside and the inside cell. It leads to faster decay of health of the battery pack. So maintaining battery temperature is a difficult task [22].

Considering the temperature variation on the battery pack, the state of estimation of temperature can be done by considering various factors like multiple points temperature variation, battery chemistry having different thermal characteristics, and computational complexity in real time.

The State of Temperature (SOT) estimation methods include direct method, model-based methods, and data-driven (DD) algorithms [23].

The direct method uses temperature sensors like thermocouple, RTD; where we can have real-time monitoring of the temperature is possible. The electrical model and thermal model of the battery pack are coupled through cell resistance. Current distribution in parallel-connected branches is determined by cell resistance, which is

calculated using the pack electrical model. The resistance model shows that cell resistance is highly influenced by temperature.

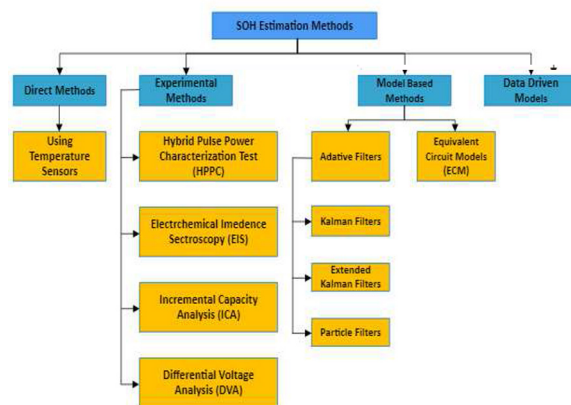


Fig. 5. Classification of Battery SOT Estimation Methods

Cell temperature is obtained by solving the pack thermal model. The battery models often exclude degradation effects and lack electro-thermal modeling at the pack level. So the models do not take interactions among cells. So there is a need to do advanced modeling [22].

V. COMMUNICATION AND NETWORKING

The efficient function of BMS depends on the exchange of information between the cell to module to pack and vice versa [24]. Now days Controller Area Network (CAN) bus, and I2C/SPI communication protocols are generally employed. It is inexpensive, can support to auto retransmission of lost messages, and works in various environmental conditions. But due to the large number of battery pack cells, the design faces several deficiencies like complication in sensing for a large number of cells, physical connection failure due to a vibratory environment, difficult troubleshooting, connection complexity for high capacity battery, and electrical loading, high maintenance cost. The total cost required for wired BMS parts like connectors, protective cases, and isolators covers almost large portion of the battery pack cost. All these problems make the BMS unit low in reliability and low in productivity [7],[25].

To solve the problems faced by the wired BMS adaption of wireless BMS has been invented. The main wireless BMS techniques are Bluetooth-based BMS, Zigbee-based BMS, Wi-Fi-based BMS, IoT-based BMS, and Cloud-based BMS. More popular are Bluetooth and Zigbee because of low power consumption. Bluetooth-based BMS is limited by a low data transfer rate, susceptibility to communication failures, and short range, making it suitable only for low-capacity batteries. Similarly, Zigbee is typically used for advanced, large battery packs but suffers from limited data transfer capabilities and security concerns. Use of cloud and IoT-based technology makes technology wider by adding Google's internet platform, so it has powerful computing and unlimited cloud support. It can be used in a micro grid. Wi-Fi-based BMS gives the advantage of scalability. So overall use of wireless BMS improves system reliability, has low weight, and reduces cost and wiring complexity[7]. The concerning factors about wireless communication are

information security, high-resolution sensors, and high-speed speed high-volume data transmission required. It also includes real-time data processing, high storage and maintenance costs, and maintaining consistent internet connectivity to the vehicle. The presence of other networks produces noise. It can lead to connection interruption, delay in connection, and transmission delay. So to implement wireless BMS technology, more reliability is needed.

VI. EXPERIMENTS AND METHODS

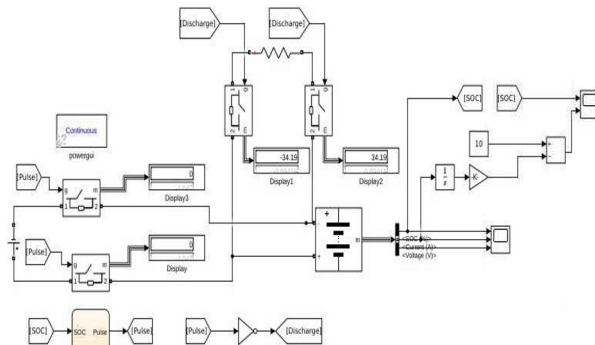


Fig. 6. Simulink Model for SOC Estimation

The Simulation is done using a Li-ion battery. The control circuit for charging and discharging is developed by using ideal switches. The battery is configured as per specifications and parameters like voltage, current, and SOC are examined. The state of charge (%SOC) is determined using the well-known Coulomb counting method and then compared with the battery's actual %SOC.

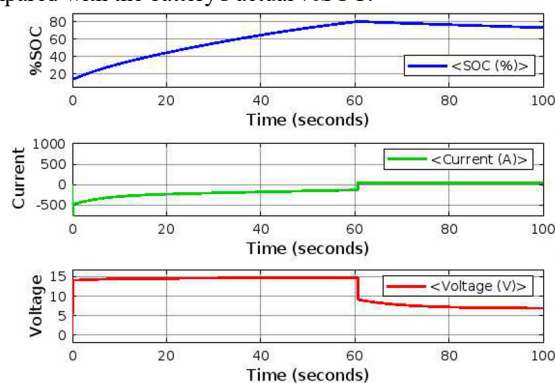


Fig. 7. Waveforms for Battery charging and discharging

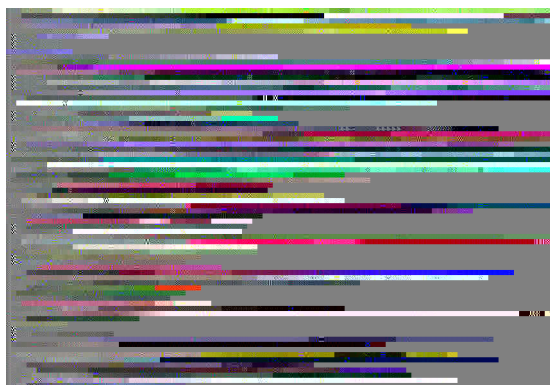


Fig. 8. Comparison of %SOC Estimation Methods

The experimental results show the charging and discharging of the battery. Initial %SOC is set to 10% and the conditional logic is developed to work the battery between 40% to 80%. The widely used Coulomb counting method is used for the estimation of %SOC and compared with the actual %SOC. The method has some drawbacks, like accuracy due to uncertainty in initial SOC.

VII. CONCLUSION

The paper has given useful insights to study the approach of the evolution of Battery Management Systems for Electric Vehicles. BMS functions depend upon various factors like battery chemistry, methods of parameter estimation, and communication and networking. The paper mainly deals with the classification of different key features regarding BMS. The findings of initial phase of the study are a) SOC, SOH and SOT is main parameters required to function a BMS b) conventional methods include direct methods, mathematical models c) for effective and real time estimation the new method of data driven method will be useful d) Network and communication plays important role in up gradation in BMS. SOC of the battery plays an important role in BMS working. The paper explores the use of the coulomb counting method and highlights drawbacks. Accurate estimation of %SOC requires error detection and correction mechanisms. Techniques such as artificial neural networks, fuzzy logic, and other AI-based algorithms have been employed to minimize these errors. As a future extension, exploring advanced data-driven methods could further enhance the effectiveness of BMS implementation.

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